RESEARCH PLAN — ADVISOR BRIEF

A Pareto-Aware Streaming HDC Classifier on Zynq with Dimension & Bit-Position Pruning

Project **1024HDC** · SystemVerilog · Vivado · ModelSim · Zynq-7020 · EMG classification

Primary venue
DATE 2027

Submission
~Sept 2026

Background. Hyperdimensional Computing (HDC) represents information as wide binary vectors on which all operations — bind (XOR), bundle (majority), permute — are bit-parallel and noise-tolerant: a strong match for FPGA edge inference. I already have a verified 1024-bit bind + permute SystemVerilog core with an AXI-Lite wrapper (`71/71` ModelSim tests passing). This proposal turns it into a complete **streaming HDC classifier** and adds an explicit research contribution that existing FPGA-HDC work has not measured: an accuracy / energy / area **Pareto study across three axes — dimension D, bundle-counter precision CNT_W, and per-bit-position pruning** — jointly evaluated on a Zynq board with shunt-measured energy on EMG.

WHAT I WILL BUILD

- **Six new RTL modules**: `item_memory`, `bundle_unit`, `pruning_mask` (Hook A), `popcount_am`, `encoder_top`, `hdc_stream_wrapper`; all parameterised on D.
- **AXI-Stream + DMA** replacing AXI-Lite to unlock streaming throughput.
- **Bare-metal C software** + a Python training pipeline that emits a per-bit pruning mask offline.
- **Bit-exact Python golden reference** + 10 k randomised co-sim vectors + dataset replay.

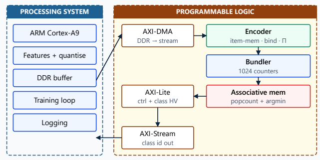
HOW I WILL EVALUATE IT

- **Primary:** UCI EMG Hand-Gesture (4 ch, 5 classes, 36 subjects) — canonical HDC benchmark.
- **Secondary:** MIT-BIH ECG arrhythmia (5 AAMI classes).
- **Baselines:** ARM-only HDC, the existing AXI-Lite path, a tiny int8 MLP, prior FPGA HDC.
- **Metrics:** accuracy, latency, energy/inference (shunt + INA219), LUT/FF/BRAM, f_{max}.

NOVELTY / CONTRIBUTION (HOOK A)

- **Three-axis Pareto study** on real Zynq hardware: $D \in \{256, 512, 1024, 2048\}$ × CNT_W ∈ {3..6} × bit-pruning ratio ∈ {0, 50, 75, 87.5 %} — jointly absent from prior FPGA-HDC work.
- **Bit-position pruning** by training-time per-bit discriminability score: the under-explored axis. Pruned bits are zeroed in the Hamming distance via a small `pruning_mask` module, saving dynamic energy ~proportionally to (1−K/D).
- **Per-bit discriminability heatmap** — visual proof that bits within a fixed D are *not* equally informative on real biosignals.

SYSTEM OVERVIEW



Five new modules (coloured) wrapped around the existing verified bind + permute core.

TARGET HEADLINE NUMBERS

≥ 92 %	2–4×	< 50 µs	10×
EMG accuracy at D=1024	Energy reduction along Pareto	End-to-end latency	vs. ARM-only baseline

TIMELINE (MAY → SEPT 2026)

May	Python golden reference; reproduce published EMG number
Jun	New RTL modules + bit-exact co-sim
Jul	AXI-Stream + DMA bring-up on board
Aug	Full experimental matrix + power measurement
Sept	Paper draft & submission to DATE 2027

Venue cascade: DATE 2027 → AICAS 2027 → FCCM 2027 → journal (TVLSI / TRETS / JETCAS).

What I would like from you. (1) Confirmation of Zynq board availability through August's power-measurement phase; (2) ~30 min of feedback in mid-July on the experimental design before I lock figures; (3) co-author review of the paper draft in early September. **Main risk:** "yet-another-HDC-paper" reviewer reflex — mitigated by leading with the streaming + AM + open-artifact contribution rather than the algorithm.